Effects of Instructional Context on Attention during Learning Activities in Children with and without ADHD

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Background

1. There exists a need to understand factors that support attention in naturalistic settings, such as classrooms, especially for children with ADHD.
2. The alpha-band oscillations show large group effects related to working memory deficits in ADHD.
3. But correlations between alpha and academic performance are low (Lenartowicz et al., 2014; Lenartowicz et al., 2019).
4. *Why?* We need to study attention deficit in natural environments.

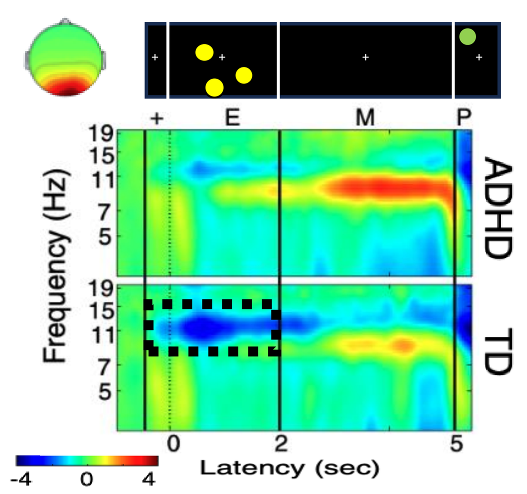


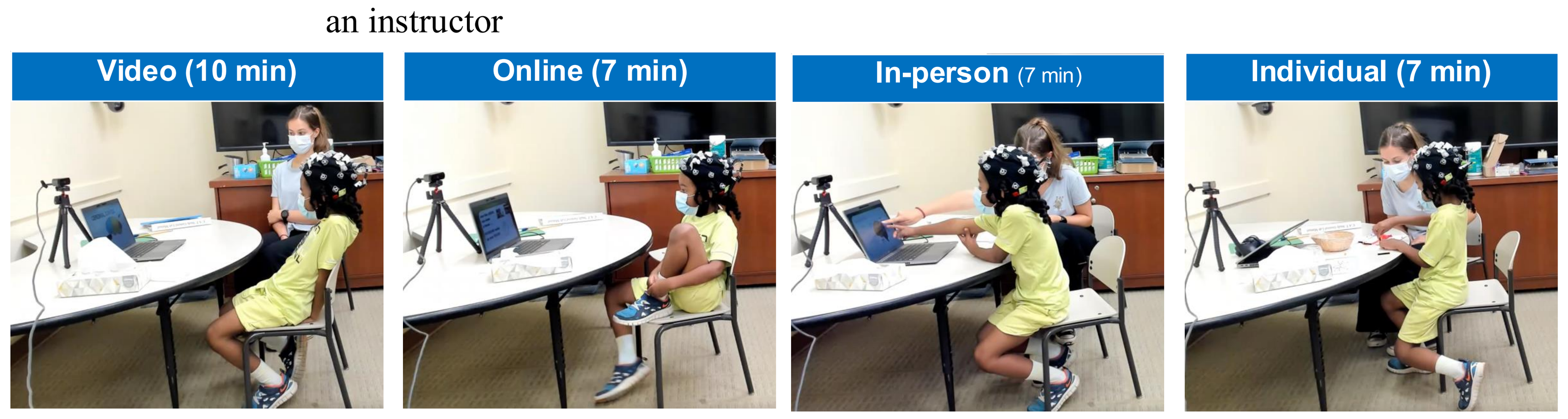
Figure 1 (left).
Alpha ERD (event-related decrease) during encoding in both TD and ADHD groups (Lenartowicz et al., 2014).

Research Questions

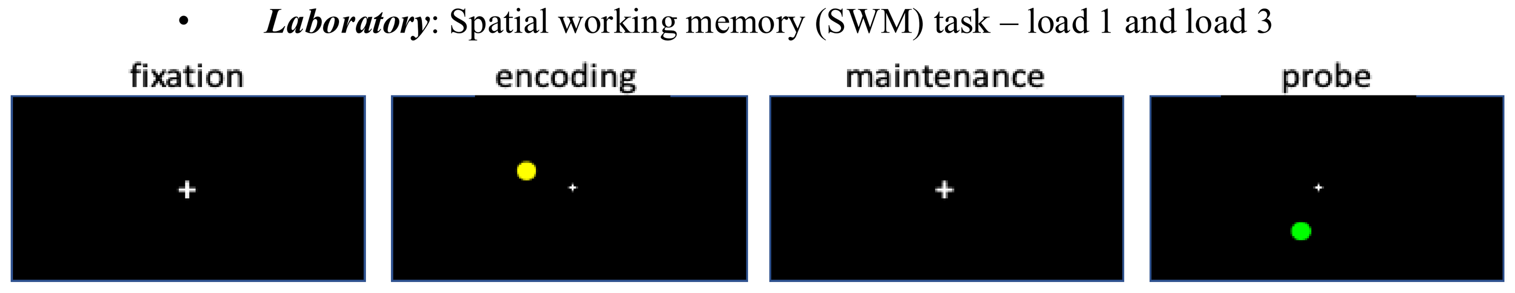
1. Does instructional context affect attention?
2. Does this differ for children with ADHD?
3. How does laboratory task performance correlate with attention measured during learning activities?

Methods

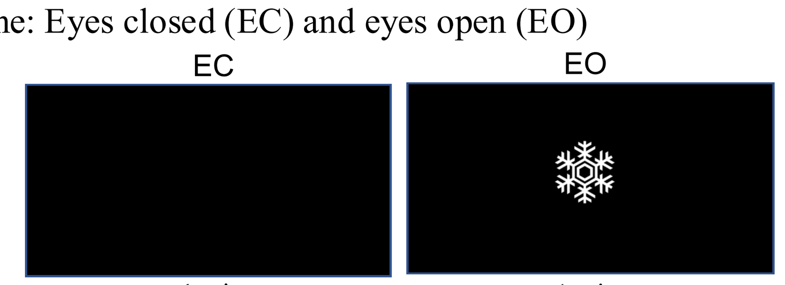
- **Participants:** 55 children (age: 8.5 years; M:F = 30:25; ADHD = 13)
- **EEG data:** Digitized at 250 Hz using Smarting MOBI with a 24-channel saline cap
- **EEG recording:**
  - *Naturalistic*: Four learning activities with different learning modalities conducted with an instructor



  - *Laboratory*: Spatial working memory (SWM) task – load 1 and load 3



  - Baseline: Eyes closed (EC) and eyes open (EO)



- **EEG analysis:** FOOOF (Python) was used to extract alpha power (9–10 Hz) from the posterior (O1, POz, O2) cluster for naturalistic, lab, and baseline EEG data. EEGLAB was used to extract alpha event-related modulation (8–12 Hz).

Results

1.1 *Naturalistic* Learning Activities

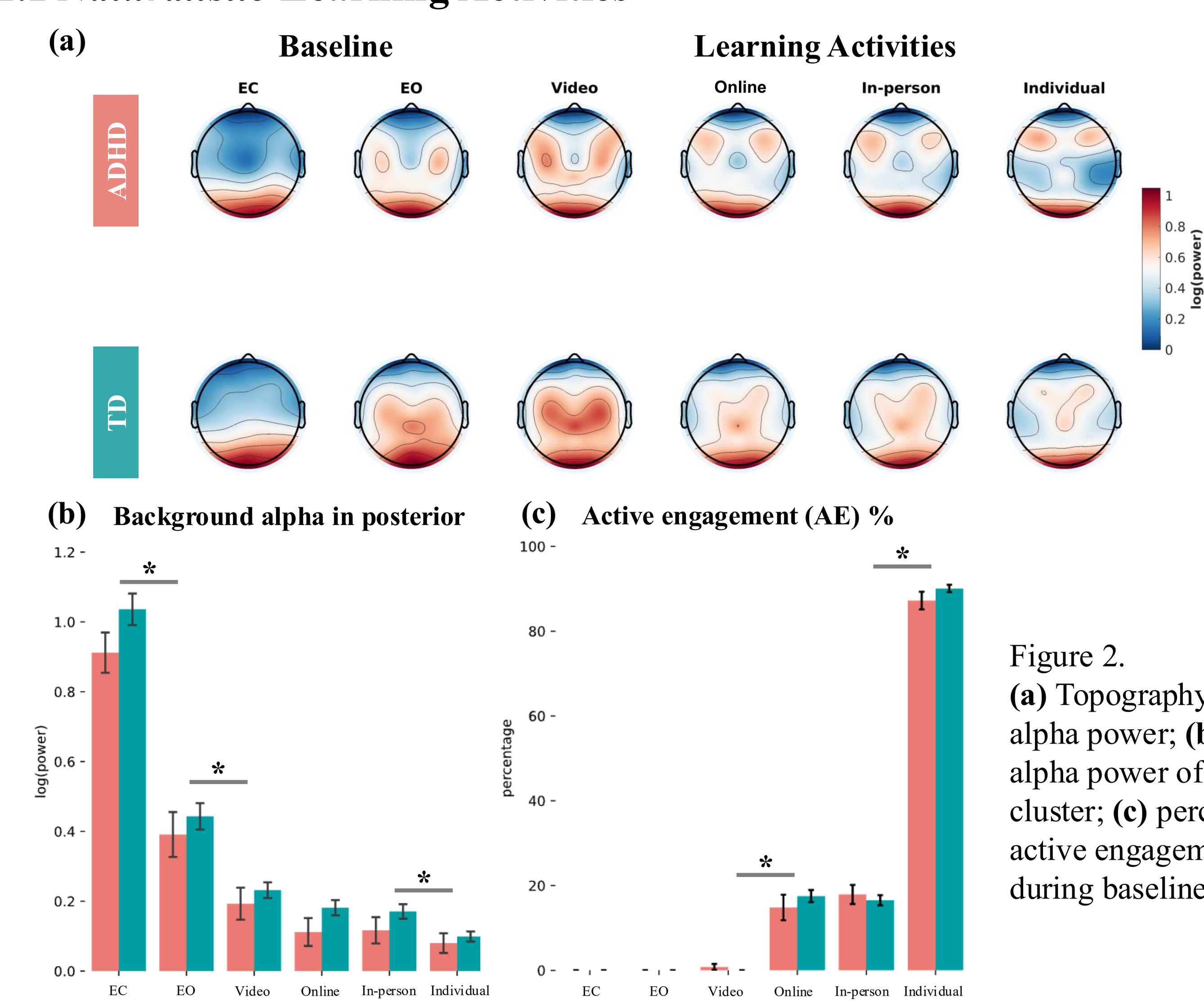


Figure 2.
(a) Topography of background alpha power; **(b)** background alpha power of the posterior cluster; **(c)** percentage of active engagement (AE%) during baseline and activities.

1.2 *Naturalistic* Learning Activities – Time Effect

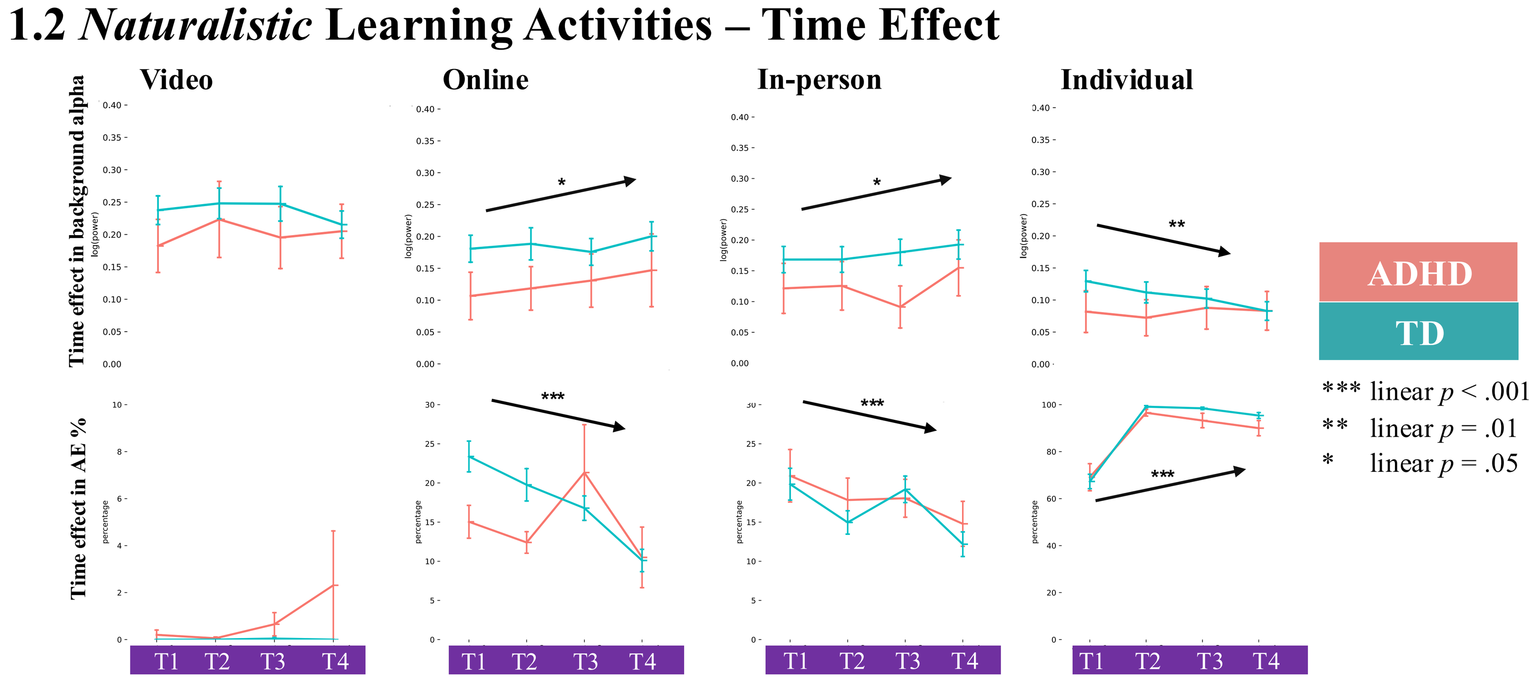
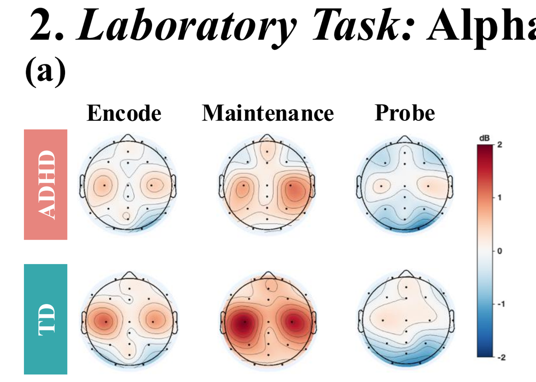


Figure 3.
Time-on-task effects for alpha power in the posterior cluster (top row) and behavioral measures (bottom row), specifically active engagement (AE%). During interactive activities (online, in-person), alpha power increased and AE% decreased, whereas the opposite pattern was observed during the student-led (individual) activity. No time-on-task effects were evident during the asynchronous video-watching activity. Error bars represent the standard error.

2. *Laboratory Task:* Alpha Modulation during SWM



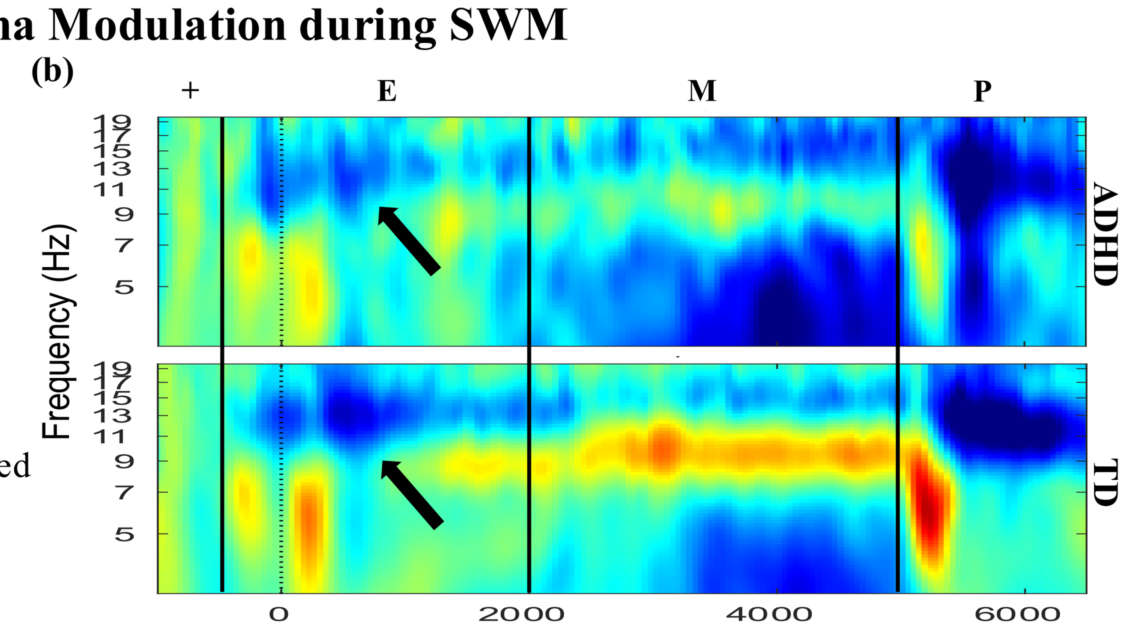
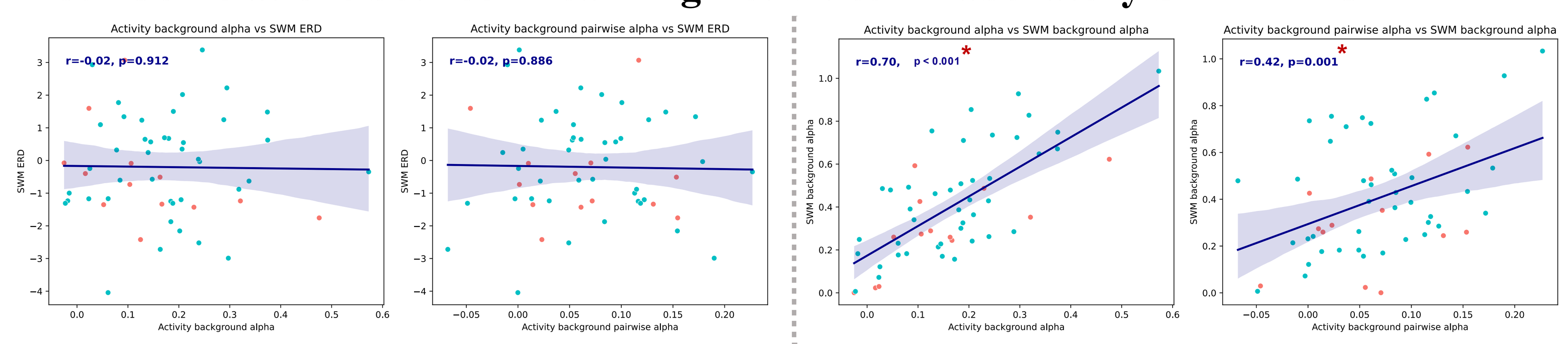


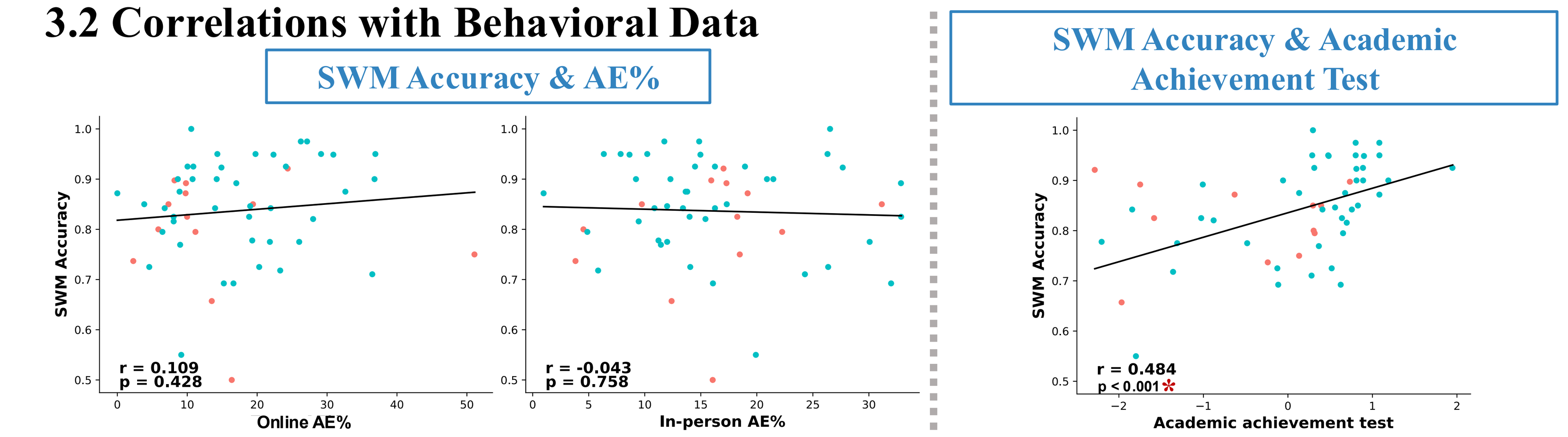
Figure 4.
(a) Topography of alpha event-related modulation during the encoding, maintenance, and probe phases; (b) Time-frequency representation of modulation in the posterior cluster, with the strongest encoding decrease observed in the beta band.

Results (cont.)

3.1 Correlations between Learning Activities – Laboratory SWM EEG



3.2 Correlations with Behavioral Data



4. Regression Analyses

Table 1. *Multiple Regression Models of Behavioral Measures on EEG Source Measures in Laboratory Setting*

	***tβ* (regression coefficient)**										**Model fit**	
	$Alpha_b$	$Alpha_e$	$Alpha_{m\perp e}$	$Alpha_{p\perp e}$	ERD_e	$ERI_{m\perp e}$	$ERD_{p\perp e}$	Age (yr)	Sex	Symptom	R^2_{adj}	*F*
Accuracy	0.881	-0.545	1.159	-0.520	1.651	**-2.289***	1.243	**4.449*****	-1.379	1.069	0.291	3.213**
RT_{mean}	-0.211	-0.945	0.989	1.280	0.777	-0.366	-0.703	**-2.190***	0.777	-0.479	0.287	1.767
RT_{CV}	-0.364	-0.602	0.878	1.271	-0.450	-1.081	-1.749	-1.542	-0.378	-0.253	-0.024	0.874
Attention	1.329	0.874	1.052	**-2.043***	1.403	**-2.028***	**2.733****	**5.582*****	0.723	0.275	0.411	4.765***
Fidgeting	-0.860	-1.209	-0.592	1.574	-1.104	1.526	**-2.489***	**-5.431*****	-1.050	-0.013	0.378	4.285***
Off-task	-1.926	1.033	-1.842	**2.059***	-1.335	**2.175***	-1.386	-1.594	1.034	-0.987	0.128	1.791

Note. Factor loadings (standardized regression coefficients, β) that exceed ±0.3 are highlighted in bold. ***p < 0.001, **p < 0.01, and *p < 0.05 indicate statistically significant regression coefficients. Alpha, background alpha power; ERD, event-related decrease; ERI, event-related increase; RT_{mean}, mean reaction time; RT_{CV}, coefficient of variation of reaction time.

Table 2. *Multiple Regression Models of Behavioral Measures on EEG Source Measures in Learning Activities*

	***tβ* (regression coefficient)**					**Model fit**	
	Alpha	ERD	Age (yr)	Sex	Symptom	R^2_{adj}	*F*
Video							
AE	**2.441***	0.405	0.474	-0.968	-1.438	0.063	1.727
PE	1.235	0.656	1.875	1.585	1.425	0.106	2.277
Fidgeting	-1.660	-0.690	-1.654	**-2.119***	-1.173	0.137	2.710*
Off-task	0.272	-0.307	-1.278	1.231	-0.695	-0.024	0.744
Online							
AE	-0.678	-0.328	0.602	-1.680	-0.128	-0.018	0.810
PE	**2.724****	0.677	2.995**	-0.580	1.629	0.271	5.009***
Fidgeting	**-3.053****	-0.302	-2.707**	1.227	-1.431	0.276	5.127***
Off-task	1.257	-1.054	**-3.661*****	1.813	-1.320	0.214	3.947**
In-person							
AE	-0.245	-0.870	0.201	0.032	-0.719	-0.071	0.281
PE	**3.587*****	0.944	**3.043****	0.304	1.516	0.327	6.247***
Fidgeting	**-2.996****	-0.647	-1.941	-0.152	-1.099	0.192	3.561**
Off-task	-1.054	0.115	**-2.595****	-0.497	-0.143	0.066	1.758
Individual							
AE	0.037	-0.067	0.374	0.076	1.537	-0.047	0.516
PE	0.648	0.023	0.234	1.262	0.803	-0.044	0.543
Fidgeting	-1.060	-0.477	-0.228	-0.431	-1.896	0.004	1.043
Off-task	-0.199	0.582	-0.772	-1.620	-1.908	0.044	1.502

Note. Factor loadings (standardized regression coefficients, β) that exceed ±0.3 are highlighted in bold. ***p < 0.001, **p < 0.01, and *p < 0.05 indicate statistically significant regression coefficients. Alpha, background alpha power; ERD, event-related decrease; AE, active engagement; PE, passive engagement.

- Table 1 shows that background alpha power appears to contribute less consistently to task outcomes than event-related measures, but shows clearer associations with video-coded behavior (attention, off-task).
- Table 2 shows that background alpha power is more consistently associated with video-coded behavioral measures than ERD, with distinct patterns observed for AE and PE across interactive (online, in-person) and student-led (individual) activities.

Conclusion

1. Instructional context has a significant effect on attention and behavioral engagement during learning activities, regardless of ADHD diagnosis.
2. Laboratory measures may be narrowly capturing visual encoding during SWM, but this is not related to the behaviors associated with attention in naturalistic activities, whereas the continuous alpha power captures background processes that are related to natural behavior.

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